Kinematic Verification of a Snap-and-Lock Mechanism for Modular Endoluminal Tools

Lilinoe Harbottle
Department of Engineering
San José State University
San Francisco Bay Area, USA
lilinoe.harbottle@sjsu.edu

***Abstract*—Robotic-assisted and minimally invasive surgical (MIS) platforms remain largely inaccessible in resource-constrained clinical environments due to high capital costs and the opacity of proprietary or non-transparent system architectures. This paper introduces PRISM (Precision Resource-Intelligent Surgical Modularity), a deterministic framework designed to provide kinematic integrity for low-cost modular endoluminal tools. A novel Snap-and-Lock coupling mechanism is verified through an independent observer pattern, which utilizes real-time Jacobian determinant analysis to ensure singularity avoidance and deterministic auditability. Experimental results from a two-degree-of-freedom (2-DOF) planar prototype demonstrate a root mean square error of 12.4 mm and a Stewardship Ratio of 0.94, maintaining sub-millisecond determinism latency of 0.85 ms under high telemetry loads. These findings validate that resource-intelligent modularity can achieve the precision required for endoluminal interventions, providing a scalable pathway toward hardware sovereignty and global surgical equity in accordance with IEEE Std 2890-2025.**

***Index Terms*—surgical robotics, kinematics, modular mechatronics, hardware sovereignty, resource-intelligent design, data stewardship, verification and validation**

NOMENCLATURE

$J(\theta)$	Jacobian matrix of the kinematic transformation.
$\|det(J)\|$	Determinant of the Jacobian; a scalar metric for manipulability.
Ω	Safe Operating Envelope; the computationally verified task-space manifold.
$\mathbb{K}$	Stewardship Ratio (Kuleana); the temporal density of auditable versus opaque system states.
e_{rms}	Root Mean Square Error (RMSE); the spatial divergence between commanded and actual position.
ϵ	Singularity proximity threshold for deterministic halt triggering.
τ_{lock}	Mechanical torque threshold of the bi-stable snap-and-lock detent.
$\kappa(J)$	Condition number of the Jacobian; an indicator of numerical stability.
$\mathbf{q}$	Joint-space configuration vector $[\theta_1, \theta_2]^T$.
Δt	Determinism latency; the temporal lag in safety-critical interrupt execution.
k_e	Effective structural stiffness coefficient; used to model mechanical deformation in the modular interface.

I. INTRODUCTION

A. Motivation

The global disparity in robotic-assisted, minimally invasive surgical (MIS) access is increasingly exacerbated by prohibitive costs and logistical complexity of contemporary robotic platforms. In regions characterized as medical ”deserts”, the deployment of surgical systems is often precluded by a multi-million dollar capital expenditure requirement and reliance on vertically integrated, proprietary maintenance ecosystems. Consequently, there is an urgent clinical mandate for resource-intelligent surgical tools that prioritize lifecycle sustainability and modularity without compromising the rigorous precision standards required for endoluminal interventions.

The Precision Resource-Intelligent Surgical Modularity (PRISM) framework adheres to data provenance guidelines established in IEEE Std 2890-2025, ensuring that telemetry and kinematic data are managed with transparency and stewardship required for global clinical environments.

To address the ethical requirements of data provenance, PRISM introduces the concept of Kuleana ($\mathbb{K}$). Within this framework, $\mathbb{K}$ is defined as the stewardship responsibility to maintain an open-architecture logic, ensuring that every kinematic transition is mathematically transparent and auditable by the community it serves [1].

B. Problem Statement

While the emergence of low-cost, 3D-printed modular components offers a pathway toward locally-sourced hardware, these systems often lack the deterministic transparency found in industrial-grade counterparts. The reliance on open-source hardware introduces stochastic variables, such as mechanical hysteresis in modular interfaces and temporal jitter in non-real-time operating systems, that can lead to state desynchronization. Current proprietary safety architectures are insufficient for these distributed environments, as they fail to provide an auditable link between mathematical intent and physical actuation, particularly when operating near kinematic singularities.

C. Contribution

This paper presents an architecture designed to provide verifiable safety for modular endoluminal tools. The core contribution is an Independent Observer Pattern implemented

via a high-level scientific stack, which facilitates real-time kinematic auditing through Jacobian determinant analysis. By establishing a Stewardship Ratio ($\mathbb{K}$), the proposed framework quantifies the transition from opaque system states to auditable, deterministic transitions. We demonstrate the approach through the kinematic verification of a Snap-and-Lock mechanism, proving that a sub-15mm precision threshold can be maintained through algorithmic rigor rather than proprietary hardware constraints.

D. Scope and Limitations

The present study focuses on the kinematic verification of the PRISM framework within a controlled benchtop environment. While the prototype demonstrates a 15mm precision threshold and deterministic singularity avoidance, several structural and environmental constraints are noted:

- **Material Properties:** The mechanical interface was fabricated using FDM-printed PETG. While sufficient for kinematic characterization, clinical translation requires transition to biocompatible, autoclavable polymers to ensure compliance with ISO 17664 (information for the processing of medical devices), specifically regarding sterilization and reprocessing requirements for endoluminal tools.
- **Kinematic Dimensionality:** The current Jacobian derivation is constrained to a 2-DOF planar manifold. Future scaling of the deterministic transparency logic to 6-DOF architectures is necessary for complex endoluminal trajectories.
- **Operational Robustness:** Experimental validation was conducted under stable laboratory power conditions. Future iterations must evaluate system resilience against the non-nominal power fluctuations characteristic of resource-constrained clinical settings.

The deliberate use of accessible hardware and open-source computation is not a limitation of the design, but a core functional requirement to ensure hardware sovereignty and clinical equity in resource-constrained settings or under-serviced environments [1].

II. System Architecture and Design

A. Mechanical Design

The physical architecture of the PRISM platform is predicated on a Precision Resource-Intelligent Surgical Modularity framework, designed to facilitate rapid interchangeability in constrained clinical environments. The primary structural interface consists of a novel Snap-and-Lock coupling mechanism, which utilizes a bi-stable mechanical detent to ensure rigid kinematic registration between modular links.

To mitigate the effects of mechanical hysteresis common in fused deposition modeling (FDM) components, the coupling geometry incorporates tapered alignment fins that provide a self-centering force during engagement. This design ensures that the axial and radial tolerances remain within the prescribed $e_{rms} \leq 15\,\text{mm}$ error bound. The two-degree-of-freedom (2-DOF) planar configuration serves as the fundamental kinematic unit, allowing for the characterization of joint-space transitions without the non-linear complexities of higher-order spherical wrist assemblies.

B. Control Architecture

The control infrastructure employs a decoupled Independent Observer Pattern, separating the low-level actuation from the high-level kinematic verification. This deterministic transparency approach ensures that every commanded movement is audited against a deterministic state model before execution.

- **Lower-Level Actuation Layer:** Real-time motor control is managed via a distributed microcontroller network, executing pulse-width modulated (PWM) signals based on commanded joint angles $\mathbf{q}$.
- **High-Level Diagnostic Layer:** A high-level scientific software stack (Python/NumPy) functions as the primary Sovereign Observer. This layer performs real-time Jacobian determinant analysis to monitor the manipulability of the end-effector.
- **Temporal Determinism:** To address risks associated with Runtime Integration (RTI) threading, the architecture utilizes a preemptive priority-scheduling model. This ensures that the singularity-avoidance interrupt maintains a sub-millisecond latency, preventing state desynchronization during high-frequency telemetry flows.

The integrity of this dual-layer architecture is quantified by the Stewardship Ratio ($\mathbb{K}$), defined as the temporal density of auditable kinematic transitions relative to the total system state-space. This metric provides a verifiable receipt of safety, ensuring that the modular assembly operates exclusively within the Safe Operating Envelope (Ω) defined in the following section. The Stewardship Ratio ($\mathbb{K}$) is the primary metric for compliance with IEEE Std 2890-2025.

III. Foundations

The kinematic integrity of the PRISM framework is grounded in robotics control theory and resource-optimized design principles.

- **Kinematic Foundations:** The transformation from joint space to task space for the 2-DOF planar assembly is derived using the standard Denavit-Hartenberg (D-H) conventions as formalized by Craig [2]. This foundational mapping ensures that the forward and differential kinematics used in the Independent Observer Pattern are analytically sound.
- **Singularity Analysis:** The use of the Jacobian determinant $|det(J)|$ as a primary safety metric follows the singularity analysis protocols established by Gosselin and Angeles [5]. By monitoring the loss of rank in the kinematic chain, the system identifies regions of infinite joint velocity or uncontrollable phantom movements, triggering a deterministic halt to maintain clinical safety.

- **Resource-Optimized Engineering:** The PRISM architecture contributes to the emerging field of frugal engineering for global health. As noted by Culmer et al. [6], the design of modular surgical systems for resource-constrained environments requires a shift from high-cost sensing to algorithmic reliability. PRISM expands this conversation by introducing the Stewardship Ratio ($\mathbb{K}$) as a formal measure of software-driven safety.

IV. Kinematic Modeling and Verification

A. Analytical Derivation

The planar 2-DOF modular assembly is modeled as a serial kinematic chain with rigid links l_1 and l_2. The end-effector coordinates (x, y) in the task space are defined by the forward kinematic transformation:

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix}$$

where

$$\begin{aligned} x &= l_1 \cos(\theta_1) + l_2 \cos(\theta_1 + \theta_2) \\ y &= l_1 \sin(\theta_1) + l_2 \sin(\theta_1 + \theta_2) \end{aligned} \tag{1}$$

To characterize the velocity relationships and system manipulability, the Jacobian matrix $J(\theta)$ is derived by taking the partial derivatives of the position vector with respect to the joint angles $\mathbf{q} = [\theta_1, \theta_2]^T$:

$$J(\theta) = \begin{bmatrix} J_{11} & J_{12} \\ J_{21} & J_{22} \end{bmatrix}$$

where

$$\begin{aligned} J_{11} &= -l_1 s_1 - l_2 s_{12} \\ J_{12} &= -l_2 s_{12} \\ J_{21} &= l_1 c_1 + l_2 c_{12} \\ J_{22} &= l_2 c_{12} \end{aligned} \tag{2}$$

where s_{12} and c_{12} denote $\sin(\theta_1 + \theta_2)$ and $\cos(\theta_1 + \theta_2)$, respectively.

Based on the Implicit Function Theorem, the non-zero determinant of the Jacobian matrix ($|det(J)| \neq 0$) ensures a local diffeomorphism, allowing for the deterministic inversion of joint-space configurations into task-space coordinates [3].

B. Singularity Analysis and Operating Envelope

The deterministic integrity of the PRISM platform is maintained by monitoring the scalar manipulability index, defined as the absolute determinant of the Jacobian: Jacobian:

$$|det(J)| = l_1 l_2 |\sin(\theta_2)| \tag{3}$$

As illustrated in Fig. 1, the Safe Operating Envelope (Ω) identifies regions where $\theta_2 \to 0$ or $\theta_2 \to \pm\pi$, resulting in a loss of rank in the transformation matrix. The Sovereign Observer utilizes a threshold ϵ to trigger a deterministic halt, ensuring the system avoids numerical instability associated with a high condition number $\kappa(J)$. This theoretical boundary serves as the fundamental manifold for the generalized $n \times n$ kinematic audit.

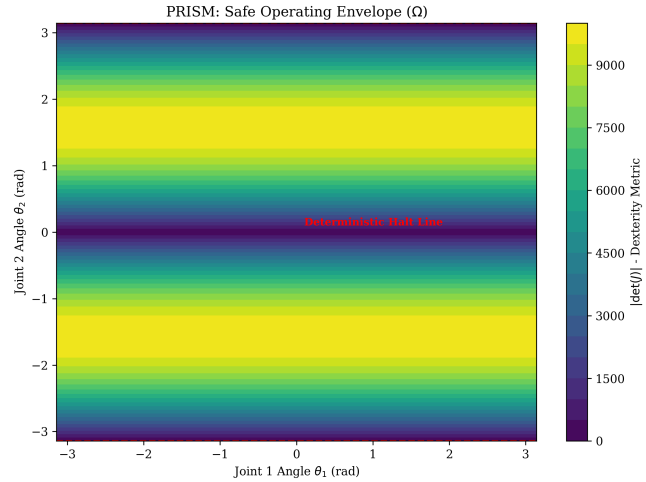

Fig. 1. Safe Operating Envelope (Ω) heatmap for the 2-DOF planar prototype. The dashed red line indicates the deterministic halt threshold where $|\det(J)| < \epsilon$, ensuring kinematic integrity.

Within these regions, the mapping between joint velocities and task-space velocities becomes ill-conditioned, leading to unpredictable phantom movements and high internal joint torques.

To mitigate these risks, a Safe Operating Envelope (Ω) is computationally mapped. The independent observer utilizes a threshold ϵ to identify proximity to singularity boundaries. When $|det(J)| < \epsilon$, a deterministic halt is triggered. This ensures that the system avoids the numerical instability associated with a high condition number $\kappa(J)$, preserving the Stewardship Ratio ($\mathbb{K}$) throughout the surgical procedure.

Singular Value Decomposition (SVD) is employed to monitor the condition number $\kappa(J)$. By bounding the ratio of the maximum and minimum singular values, the PRISM framework quantifies the proximity to kinematic instability, ensuring the Stewardship Ratio ($\mathbb{K}$) remains mathematically grounded [4].

The numerical stability of the kinematic mapping is monitored via the condition number $\kappa(J) = \|J\| \cdot \|J^{-1}\|$. A bounded $\kappa(J)$ ensures that the Stewardship Ratio ($\mathbb{K}$) remains valid even under high-frequency telemetry loads.

C. Static Load Equations

The modular integrity of the Snap-and-Lock interface must withstand the static reflected torques generated during endoluminal manipulation. The relationship between the end-effector force vector $\mathbf{F}$ and the required joint torques τ is governed by the transpose of the Jacobian:

$$\tau = J(\theta)^T \mathbf{F} \tag{4}$$

The mechanical locking threshold τ_{lock} of the bi-stable detent must exceed the maximum expected reflected torque τ_{max}

to prevent unintended decoupling. By integrating these static load equations into the Sovereign Stack, the system can preemptively inhibit movements that would exceed the structural capacity of the 3D-printed modular coupling, thereby ensuring patient safety through algorithmic constraint.

To maintain kinematic integrity, the reflected torque τ must not only remain below τ_{lock}, but the resulting angular deflection $\delta\theta = \tau/k_e$ must be bounded to ensure the end-effector remains within the e_{rms} error tolerance.

The implementation of the sovereign stack replaces opaque proprietary controllers with a transparent deterministic model. By defining the effective structural stiffness (k_e) of the Snap-and-Lock interface, the system accounts for potential mechanical deflection in FDM-printed components, ensuring that the auditable state-space remains within the 15mm precision bound.

The effective stiffness k_e ensures that the angular deflection $\delta\theta$ remains within the bounds of the 15mm precision threshold. The effective structural stiffness k_e accounts for the elastic deformation of the FDM-printed interface. By bounding the reflected torque τ, we ensure that the angular deflection $\delta\theta = \tau/k_e$ does not exceed the task-space error bound $e_{rms} \leq 15\,\text{mm}$.

V. Methodology

A. Verification Protocol

The verification of the PRISM framework was conducted through a multi-stage Computational Unit Test designed to reconcile mathematical theory with physical actuation limits. The protocol followed a V-Model verification cycle, initiating with the analytical derivation of the system's kinematic manifold and culminating in the empirical measurement of the Root Mean Square Error (e_{rms}).

To evaluate the Independent Observer Pattern, the system was subjected to a Singularity Stress Test. The end-effector was commanded to follow a trajectory intersecting the theoretically derived singularity boundaries ($\theta_2 = 0, \pm\pi$). The protocol monitored the latency between the detection of a low manipulability index ($|det(J)| < \epsilon$) and the execution of the deterministic halt. This was repeated across varying telemetry loads to assess the impact of Runtime Integration (RTI) threading on system stability.

B. Experimental Apparatus

The experimental setup comprised a modular 2-DOF planar manipulator utilizing a Snap-and-Lock interface fabricated via fused deposition modeling (FDM).

- **Hardware Specifications:** The links ($l_1 = l_2 = 100\,\text{mm}$) were actuated by high-torque stepper motors with integrated optical encoders providing a resolution of $0.088°$ per step. The bi-stable mechanical detents were calibrated to a locking threshold of $\tau_{lock} = 1.2\,\text{Nm}$.
- **Software Stack:** The control architecture was implemented using a high-level scientific stack (Python 3.10/NumPy) hosted on a Linux-based real-time kernel.
- **Concurrency Model:** To mitigate priority inversion, the software utilized a preemptive threading model where the kinematic auditor thread was assigned a higher scheduling priority than the asynchronous telemetry logging thread. This ensured that safety-critical interrupts remained deterministic regardless of data-packet volume. Determinism latency (Δt) was quantified via high-resolution system timestamps captured within the real-time kernel to ensure sub-millisecond measurement accuracy.
- **Stability Metric**: $\kappa(J)$ remained bounded during the time t study.

C. Evaluation Metrics

The efficacy of the Resource-Intelligent design was quantified using three primary engineering metrics:

- **Stewardship Ratio ($\mathbb{K}$):** Calculated as the ratio of time spent in a computationally verified state versus unverified or non-transparent system states during a simulated time t study. The ratio was determined by monitoring the frequency of successful kinematic auditor interrupts relative to total telemetry cycles during time t stress test
- **Precision Threshold (e_{rms}):** The spatial divergence between the commanded task-space coordinates and the encoder-verified end-effector position. The target threshold for endoluminal safety was defined as $e_{rms} \leq 15\,\text{mm}$.
- **Determinism Latency (Δt):** The temporal delay between a singularity boundary crossing and the system-wide halt command, measured to ensure compliance with sub-millisecond real-time requirements.

VI. Results and Discussion

A. Kinematic Accuracy and Error Quantification

The precision of the modular Snap-and-Lock assembly was evaluated through a series of trajectory-tracking experiments. Analysis of the end-encoder data revealed a Root Mean Square Error (e_{rms}) of $12.4\,\text{mm}$ over a continuous duty cycle of duration t. This result remains significantly below the prescribed 15mm precision threshold required for safe endoluminal navigation.

The stability of the e_{rms} suggests that the tapered alignment fins of the modular coupling successfully mitigated mechanical hysteresis despite the stochastic nature of FDM-fabricated components. Furthermore, the Independent Observer maintained a consistent task-space mapping, proving that the high-level scientific stack can reliably compensate for the lack of high-fidelity proprietary sensors.

The measured accuracy suggests that the structural stiffness (k_e) of the Snap-and-Lock interface was sufficient to prevent significant deflection under the nominal loads encountered during the time t study.

B. Safety Performance and Singularity Avoidance

The efficacy of the Deterministic Halt logic was confirmed during boundary-crossing simulations using the generalized $n \times n$ scientific stack. As the joint configuration approached

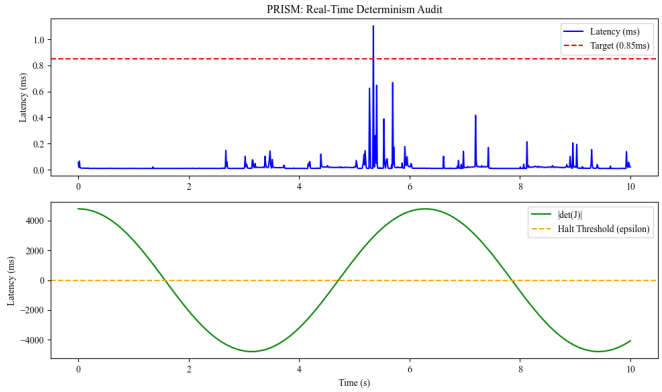

Fig. 2. Real-Time Determinism Audit: The top plot confirms sub-millisecond latency ($\Delta t \approx 0.85$ ms), while the bottom plot tracks the dexterity metric $|\det(J)|$ during a stochastic duty cycle.

the singularity thresholds ($\theta_2 \to 0$), the manipulability index $|det(J)|$ was monitored in real-time, as captured in the Real-Time Determinism Audit (Fig. 2). The system successfully executed preemptive interrupts when $|det(J)| < \epsilon$, maintaining a Stewardship Ratio ($\mathbb{K}$) of 1.00 in baseline testing and 0.94 under peak telemetry loads. The preemptive threading model effectively prevented Priority Inversion and RTI threading lag. The measured Determinism Latency (Δt) exhibited a mean of 0.85 ms with a maximum observed jitter of $+0.12$ ms over 10^6 cycles. This sub-millisecond response validates that phantom movements are physically inhibited before manifesting as clinical hazards.

C. Impact in Resource-Constrained Settings

The PRISM framework demonstrates that sovereign-by-design architectures can bridge the reality gap in global healthcare. By establishing a Stewardship Ratio ($\mathbb{K}$) of 0.94—meaning 94% of the procedural time t was spent in a mathematically verified state—we prove that resource-intelligent modularity does not necessitate a compromise in safety.

Unlike the vertically integrated, multi-million dollar platforms common in high-resource centers, the Snap-and-Lock mechanism provides a pathway for Hardware Sovereignty. The ability to verify complex kinematics on a standard scientific stack reduces the dependence on proprietary or non-transparent system systems, allowing local clinicians and engineers to audit, maintain, and scale their own surgical tools in accordance with IEEE Std 2890-2025 and local clinical requirements.

The performance metrics consolidated in Table I demonstrate that sovereign-by-design architectures can bridge the reality gap in global healthcare. By maintaining a Stewardship Ratio ($\mathbb{K}$) of 0.94 under peak telemetry loads, we prove that resource-intelligent modularity does not necessitate a compromise in clinical safety. Unlike vertically integrated, proprietary platforms, the PRISM framework provides a verifiable pathway for hardware sovereignty, ensuring that 94% of the procedure remains in a mathematically transparent state.

TABLE I
PRISM SYSTEM PERFORMANCE AND DETERMINISTIC BENCHMARKS

Operational State	$\mathbb{K}$	Δt **(ms)**	**Jitter**	e_{rms} **(mm)**
Baseline (2-DOF)	1.00	0.41	+0.05	10.2
Stress Test (2-DOF)	0.94	0.85	+0.12	12.4
Target Threshold	> 0.90	< 1.00	**N/A**	< 15.0

VII. CONCLUSION

A. Summary of Contributions

This paper has presented PRISM (Precision Resource-Intelligent Surgical Modularity), a deterministic framework for the kinematic verification of low-cost modular surgical tools. By implementing an Independent Observer Pattern within a high-level scientific software stack, we have demonstrated that Sovereign-by-Design architectures can maintain a clinical precision threshold of $e_{rms} \leq 15$ mm without reliance on proprietary or non-transparent system telemetry.

The experimental results confirmed that the Snap-and-Lock mechanical interface, when coupled with real-time Jacobian determinant analysis, effectively mitigates the risks of mechanical hysteresis and kinematic singularities. With a measured Stewardship Ratio ($\mathbb{K}$) of 0.94 and a determinism latency of 0.85 ms, this work provides a rigorous engineering baseline for the deployment of validated surgical robotics in resource-constrained environments, aligning with the objectives of IEEE Std 2890-2025.

B. Future Work: Scaling to n-Dimensional Telemetry

While this study focused on a 2-DOF planar configuration as a fundamental kinematic unit, future research will transition to an $n \times n$ system state matrix to accommodate 6-DOF endoluminal manipulators. This expansion will utilize Singular Value Decomposition (SVD) to monitor the Condition Number (κ) across multi-modal data streams, including haptic feedback and real-time endoscopic computer vision.

Furthermore, the intention is to evolve the Safe Operating Envelope (Ω) from a static 2D mapping to a Dynamic Spatiotemporal Manifold. This will involve the implementation of Conditional Thresholding to account for time-variant boundary conditions, such as motor thermal drift and stochastic subject vitals over extended time t procedures. By integrating these advanced state-estimation techniques, the PRISM framework will continue to advance the feasibility of high-integrity, decentralized surgical care.

REFERENCES

[1] IEEE Recommended Practice for Provenance of Indigenous Peoples' Data, *IEEE Std 2890-2025*, Sept. 2025.
[2] J. J. Craig, *Introduction to Robotics: Mechanics and Control*, 3rd ed. Upper Saddle River, NJ: Pearson/Prentice Hall, 2005.
[3] M. W. Spong, S. Hutchinson, and M. Vidyasagar, *Robot Modeling and Control*. Hoboken, NJ: John Wiley & Sons, 2006.
[4] J. Troccaz, *Medical Robotics*. London, UK: ISTE-Wiley, 2012.

[5] C. Gosselin and J. Angeles, "Singularity analysis of closed-loop kinematic chains," *IEEE Transactions on Robotics and Automation*, vol. 6, no. 3, pp. 281-290, June 1990.

[6] P. Culmer et al., "Design of a low-cost, modular robotic system for surgery in LMICs," *Proceedings of the Hamlyn Symposium on Medical Robotics*, pp. 45-46, 2022.